AI-DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM



R26 - IT - 102

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PROPOSAL FEEDBACK AND IMPROVEMENTS

Proposal Comment	How We Addressed It
Research gap must be clearer	Defined the gap as no integrated IoT + ML + AI + robotic harvesting system for cinnamon
Individual components must be separated	Divided project into IoT, ML, and AI robotic harvesting components
Need practical prototype evidence	Added dashboards, prediction output, AI vision flow, and robotic machine control logic
Need better integration	Created workflow: IoT data → ML prediction → AI vision → robotic machine harvesting
Need commercialization	Added target customers, value proposition, and revenue model



PROBLEM, IMPACT AND BENEFICIARIES



Problem:

- Cinnamon cultivation relies heavily on manual experience.
- No real-time monitoring of plant growth.
- Environmental conditions such as humidity, light, and soil moisture vary.
- Diseases are detected too late.
- Fmachineers cannot accurately predict harvest readiness.
- Cinnamon bark extraction is manual and inefficient.



PROBLEM, IMPACT AND BENEFICIARIES

Impact:

- Low productivity
- Inconsistent bark quality
- High labour cost
- Delayed harvesting decisions





PROBLEM, IMPACT AND BENEFICIARIES



Beneficiaries:

- Cinnamon fmachineers
- Plantation managers
- Agricultural researchers
- Cinnamon export companies
- Agricultural technology companies
- Government agriculture sector



RESEARCH GAP AND PROPOSED SOLUTION



- Captures cinnamon bark/stem image using camera or image upload.
- Uses computer vision to detect bark boundaries.
- Calculates target coordinates for harvesting.
- Sends movement instructions to robotic arm.
- Adds safety confirmation before robotic arm action.



OVERALL SYSTEM DESIGN EXCELLENCE



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AI-GUIDED ROBOTIC MACHINE FOR AUTOMATED CINNAMON BARK HARVESTING

- Detect harvest-ready bark.
- Identify cinnamon bark peeling boundaries.
- Calculate harvesting coordinates.
- Control robotic machine movements.
- Automate cinnamon bark harvesting process.





HOW SOLUTION ADDRESSES SUB-PROBLEM / PROTOTYPE



- Captures cinnamon bark/stem image using camera or image upload.
- Uses computer vision to detect bark boundaries.
- Calculates target coordinates for harvesting.
- Sends movement instructions to robotic arm.
- Adds safety confirmation before robotic arm action.



USER REQUIREMENTS ADDRESSED

- Farmers need faster bark harvesting.
- Plantation owners need consistent peeling quality.
- Users need reduced labour dependency.
- Operators need safety control /manual override.







DESIGN EXCELLENCE / CONTRIBUTION

- AI vision-based bark boundary detection.
- Coordinate generation for robotic arm movement.
- Integration with ML harvest-readiness output.
- Cinnamon-specific robotic harvesting workflow.
- Safety controls such as Start, Stop, Emergency Stop, and Manual Override.





CURRENT PROGRESS

Component	Current PP1 Progress	Evidence
IoT	Sensor selection, growth chamber design, dashboard prototype	Sensor dashboard / hardware plan
ML	Dataset preparation, model workflow, prediction prototype	Model output / charts
AI Robotics	Image upload, bark detection logic, robotic movement workflow	Detection output / control UI
Integration	IoT → ML → AI → Robotic arm workflow	Architecture diagram



RISK MITIGATION

Risk	Mitigation
Real robotic arm not fully completed	Use simulation / prototype control UI for PP1
Lack of real cinnamon dataset	Use collected sample images + public agriculture image datasets
Sensor delay or hardware issues	Use simulated sensor data until hardware is stable
ML accuracy may be low initially	Improve dataset size and model tuning
Safety risk in robotic arm movement	Add manual override and emergency stop



COMMERCIALIZATION & SUSTAINABILITY

Customer Persona

- Medium & large cinnamon plantation owners
- Cinnamon export companies
- Smart agriculture companies
- Agricultural research institutions

Market Opportunity

- Sri Lankan cinnamon cultivation and export industry
- Commercial plantations needing labour-saving automation
- Precision agriculture and smart farming market

User Benefits

- Better monitoring and decision-making
- Reduced manual labour dependency
- Improved cinnamon bark quality
- Increased harvesting efficiency



COMMERCIALIZATION & SUSTAINABILITY

Business Value & Cost Recovery



What is Unique in Our Solution

- Cinnamon-specific IoT + ML + AI + robotic harvesting system
- Predicts harvest readiness before harvesting
- Detects bark boundaries for automated peeling
- Combines monitoring, prediction, and harvesting in one platform

How Cost Will Be Recovered

- Sell hardware + software package
- Monthly dashboard/cloud subscription
- Maintenance and calibration service
- Custom robotic arm installation for plantations
- Data analytics package for large plantations





THANK YOU!